

The Impact of Short-term Intensity and Tactical Behavior on Goal-Scoring Probability: A Machine Learning Approach

Jakub Piotrowski

Jakub Szulc

Mateusz Jastrzebski

Polish-Japanese Academy of Information Technology

April 26, 2026

Abstract

We study rare-event prediction of whether a player scores after a 15-minute match checkpoint. The dataset contains 3,486 checkpoint rows (203 positives, 5.82%) across 31 fixtures and is enriched with event logs (passes, runs, shots, pressure actions). We use fold-safe temporal feature engineering, grouped cross-validation by fixture, and a stacking ensemble over out-of-fold probabilities from logistic regression, Extra Trees, Random Forest, XGBoost, and LightGBM. In the refreshed run (**fast** preset, 3 grouped folds, no XGBoost HPO, AutoGluon holdout enabled), the strongest single OOF PR-AUC is Random Forest (0.120), while stacking attains the highest ROC-AUC (0.682) and best OOF Brier score (0.054). Results confirm that short-term and cumulative in-match behaviors carry measurable signal, but absolute minority-class separability remains limited, consistent with football’s low-scoring stochastic nature.

1 Introduction

Modern football analytics can explain action value well, but forecasting an individual player’s scoring event from in-match state alone remains difficult. This work focuses on *checkpoint-time* prediction: for each player appearance at regular checkpoints, estimate the probability of scoring later in the match using only information available up to that checkpoint.

The primary contribution is a fully leakage-aware pipeline: temporal filtering by absolute match time, grouped validation by fixture, fold-local feature fitting/encoding, and explicit exclusion of future-known substitution fields from the model feature set.

2 Data and Problem Setup

2.1 Raw data

Table 1: Dataset overview (raw CSVs).

Dataset	Rows	Columns	Unique player appearances	Role
Checkpoints	3,486	33	869	Main modeling table
Passes	29,795	7	959	Passing context
Runs	35,133	10	968	Physical context
Shots	780	12	453	Shooting context
Pressure	12,185	10	923	Pressed-action context

Target variable: `scored_after`. In the training-ready checkpoint table: 203 positives and 3,283 negatives (5.82% positive rate).

2.2 Exploratory distributions

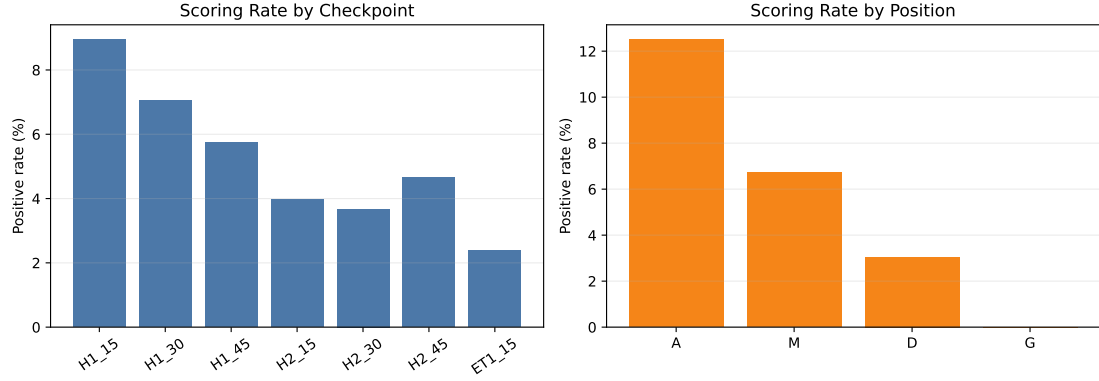


Figure 1: Scoring rate by checkpoint and position.

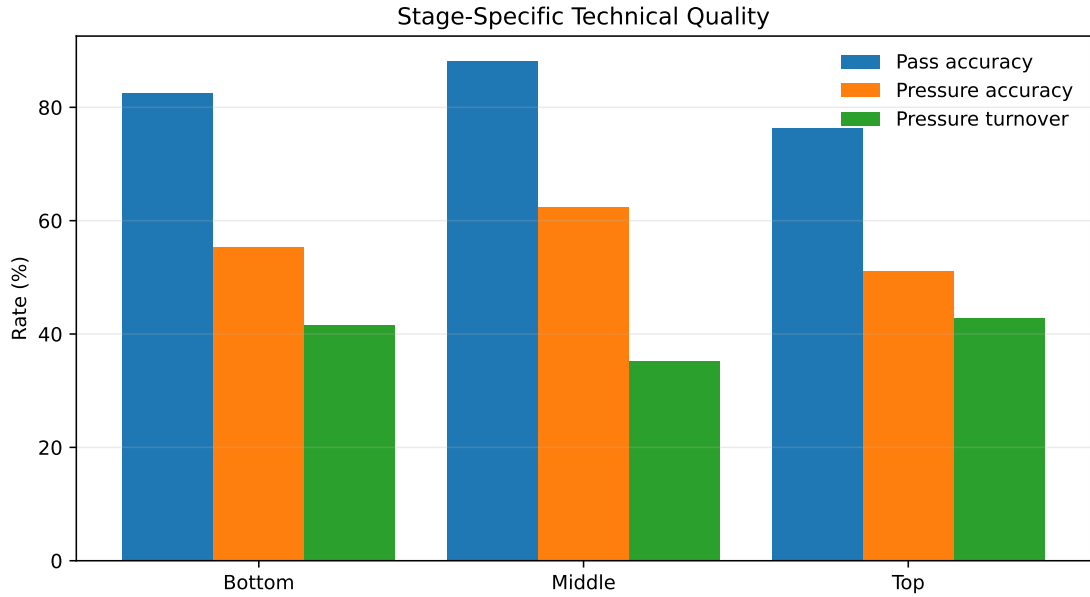


Figure 2: Pass accuracy and pressure outcomes by pitch stage.

3 Preprocessing and Leakage Controls

3.1 Training-row validation

Before feature modeling, checkpoint rows are validated with temporal leakage checks and on-pitch filtering:

- temporal boundaries are validated against event logs;
- rows where `is_on_pitch` is false are dropped;
- training aborts if no valid rows remain.

3.2 Fold-safe temporal features

Features are built inside each CV fold via `TemporalFeatureBuilder`:

- per-fold fit on training checkpoints only;
- transform applied separately to train/validation partitions;
- event filtering uses absolute match minute:

$$\text{minute_in} \leq \text{event_abs_minute} \leq \text{checkpoint_abs_minute}$$

with period offsets to avoid half-overlap artifacts.

3.3 Model matrix construction

For each fold and feature subset:

- categorical columns: missing $\rightarrow$ "missing", rare-level pooling (min count/share), one-hot encoding with train-to-val column alignment;
- additional category statistics: frequency encoding for all categoricals, plus smoothed target-rate encoding (smoothing=20) for selected columns (`position`, `formation`);
- numeric columns: coercion to numeric and median imputation fit on train only.

3.4 Excluded fields and feature groups

The modeling column selector excludes identifiers, target, and known future-leaking substitution fields (`minute_out`, `subbed`), plus `jersey_number`. The run used 241 columns in the `all_features` group after deterministic redundancy filtering.

4 Modeling Setup

4.1 Validation protocol

We use `StratifiedGroupKFold` grouped by `fixture_id`. In this run: 3 folds (`fast` preset).

4.2 Base models and stacking

Main all-feature OOF benchmark includes:

- Logistic Regression (balanced),
- Extra Trees,
- Random Forest,
- XGBoost,
- LightGBM.

Stacking trains a meta-level balanced logistic model on base-model OOF probabilities (with calibration fallback path). Fold thresholds for class predictions are selected by F1 on training fold data.

4.3 AutoGluon benchmark

An additional `autogluon_holdout` row is produced from a single train/validation split, not from full OOF across all 3,486 rows. It is useful as a complementary reference but not directly comparable with full OOF lines.

5 Results

5.1 OOF model performance

Table 2: All-features model comparison (OOF for main models, holdout for AutoGluon).

Model	n	ROC-AUC	PR-AUC	Brier
stacked	3486	0.682	0.114	0.054
random_forest	3486	0.679	0.120	0.058
extra_trees	3486	0.674	0.114	0.061
lightgbm	3486	0.656	0.114	0.059
xgboost	3486	0.650	0.107	0.060
logistic	3486	0.580	0.078	0.186
autogluon_holdout	1096	0.661	0.080	0.047

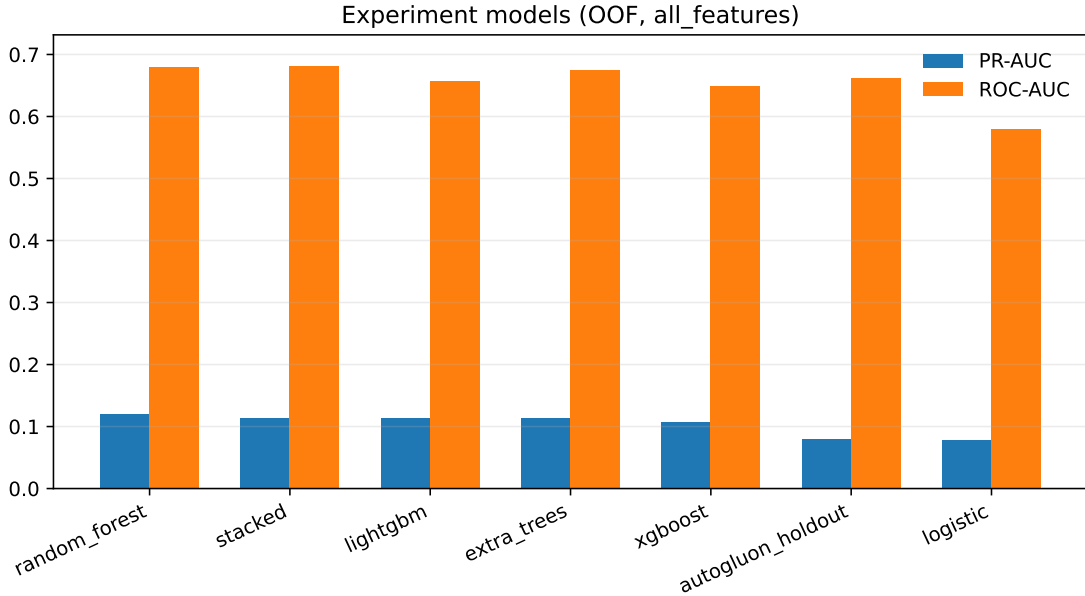


Figure 3: Model ranking plot generated from experiment exports.

5.2 Ablation view (XGBoost feature families)

Table 3: XGBoost ablations from `metrics.csv`.

Feature set	ROC-AUC	PR-AUC	Brier
base_checkpoint	0.647	0.114	0.060
base_plus_passing	0.648	0.111	0.060
base_plus_pressure	0.642	0.108	0.060
cumulative_only	0.631	0.108	0.063
all_features	0.650	0.107	0.060
recent_only	0.630	0.094	0.062
sprints_shots	0.615	0.090	0.072

external_context	0.556	0.071	0.134
------------------	-------	-------	-------

5.3 Feature-level interpretation

Most stable patterns across correlation and model-importance exports point to:

- positional/territorial effects (`is_attacker`, pass-stage shares),
- recent intensity signals (`recent_hsr_ratio`, distance ratio),
- pressure-context features (top-share / quality aggregates).

5.4 Ensemble diagnostics

Scope note. This package focuses on fast, reproducible diagnostics from OOF predictions (PR curves and meta-weights). Full SHAP/permutation analysis is intentionally excluded from the fast scan path and should be run in `full` mode when deeper attribution is needed.

6 Discussion

The pipeline consistently extracts signal above base rate, but absolute PR-AUC remains low, reflecting rare-event difficulty and noisy football outcomes. In this run, Random Forest gives the strongest minority ranking (PR-AUC), while stacking improves ROC-AUC and Brier calibration relative to individual tree models. The model family differences and meta-weights indicate complementary behavior between base learners rather than one universally dominant predictor.

7 Conclusion

Short-term and cumulative in-match behaviors provide valid predictive signal for post-checkpoint scoring probability, but are insufficient for high-certainty forecasting in isolation. The strongest practical path forward is hybridization with macro context (player history, team/opponent strength, richer game-state variables) while preserving strict grouped temporal validation.

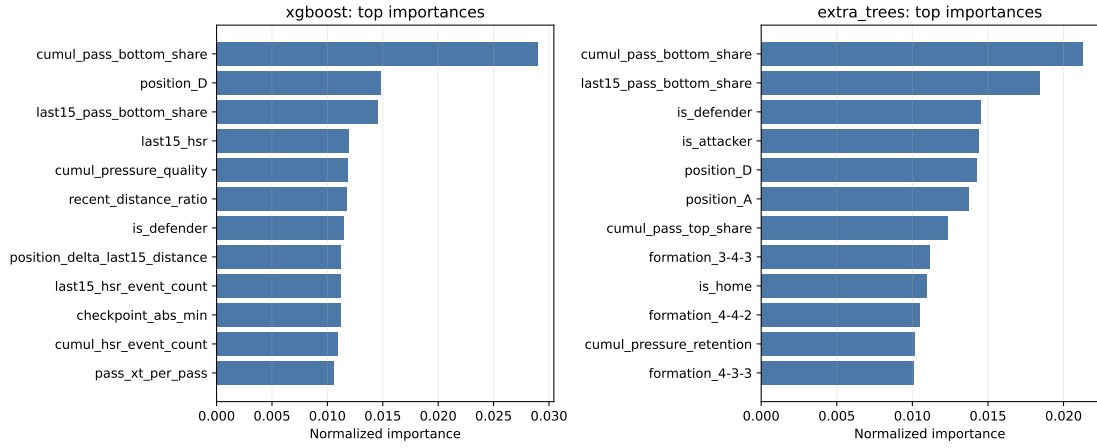


Figure 4: Normalized feature importance plot (from report figure export).

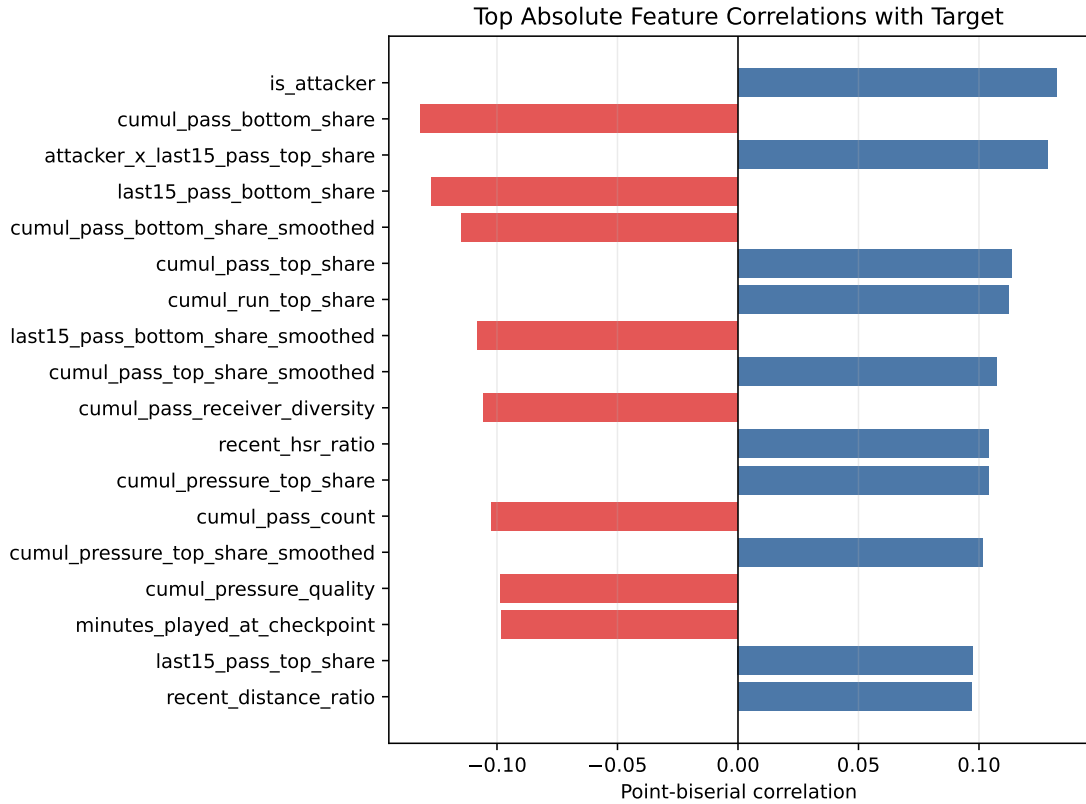


Figure 5: Top absolute correlations with `scored_after`.

Table 4: Diagnostic meta-model standardized coefficients (from explainability scan).

Base prediction	Standardized weight
random_forest_all_features	0.585
xgboost_all_features	-0.223
extra_trees_all_features	0.219
logistic_all_features	-0.204
lightgbm_all_features	0.160

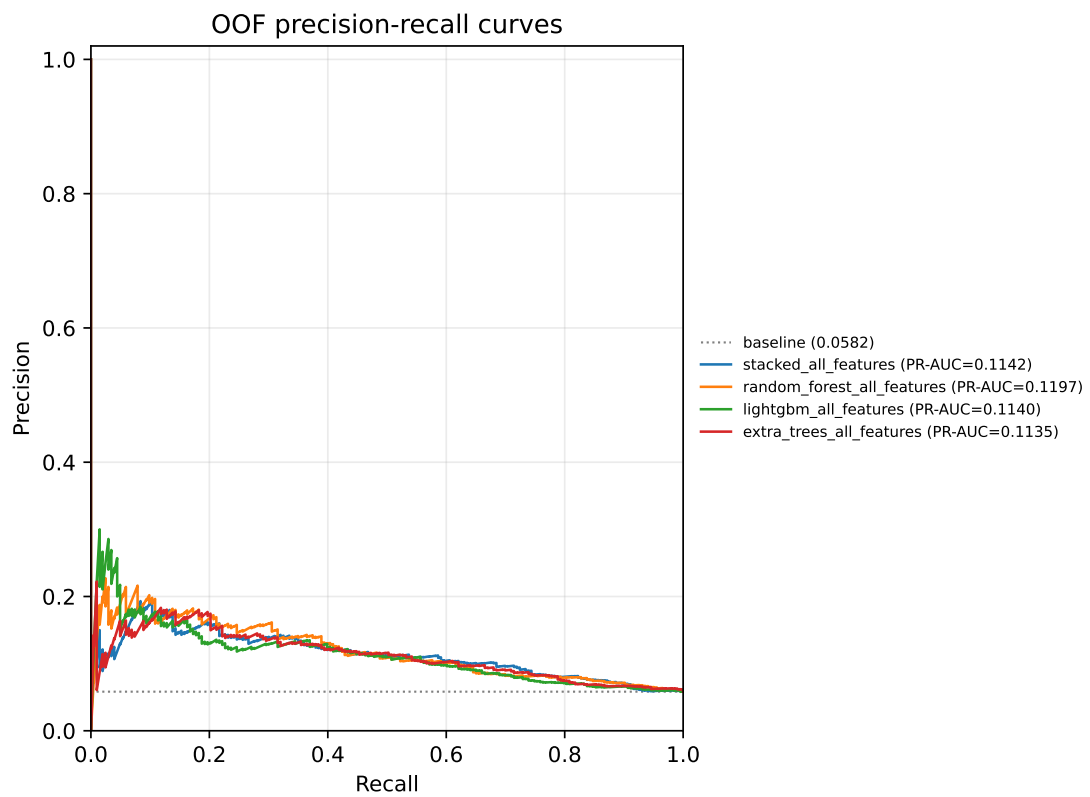


Figure 6: Precision-recall curves for OOF model probabilities with an external legend to keep traces unobstructed.

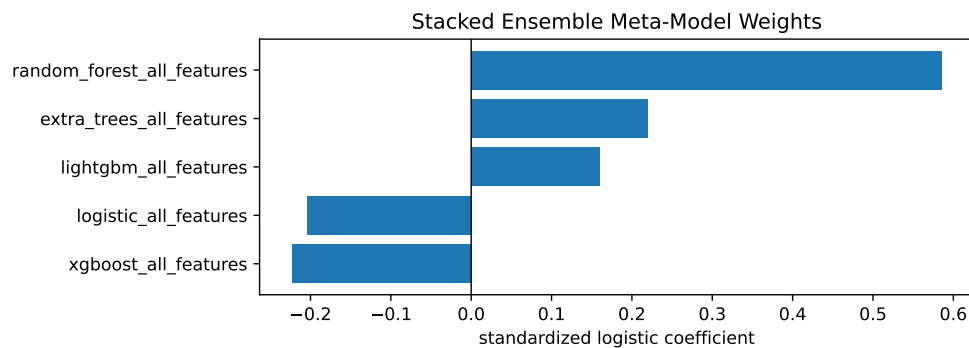


Figure 7: Post-hoc standardized weights of base-model OOF columns in a diagnostic meta-logit fit.